

Trace the solution

$$\begin{array}{c}
 (A_1 \otimes A_2) ((A_3 \otimes A_4) A_5) \\
 \begin{array}{ccccc}
 4 \times 10 & 10 \times 3 & 3 \times 12 & 12 \times 20 & 20 \times 7 \\
 p_0 \ p_1 & p_1 \ p_2 & p_2 \ p_3 & p_3 \ p_4 & p_4 \ p_5
 \end{array}
 \end{array}$$

$k=2$
 $M[1,5] = \underline{M[1,2]} + \underline{M[3,5]} + p_0 p_2 p_5$

$k=4$ ←
 $M[3,5] = M[3,4] + M[5,5] + p_2 p_4 p_5$

Edit Distance

Edit distance

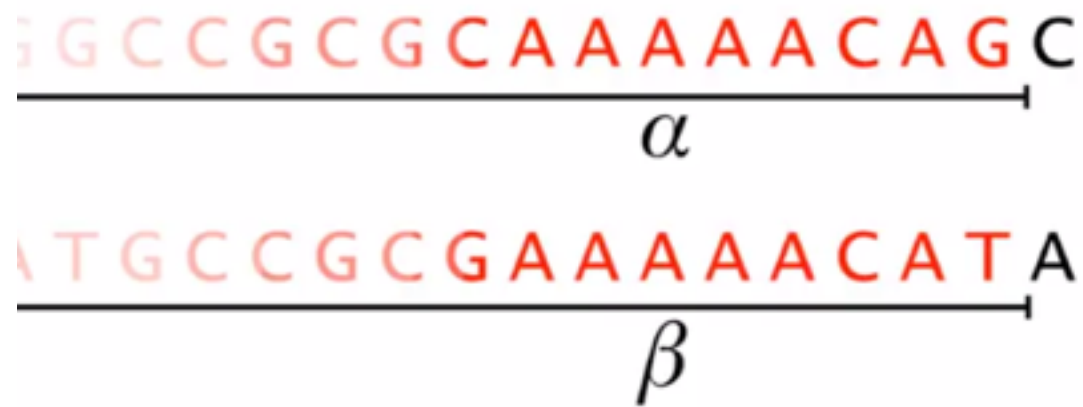
For X, Y where $|X| = |Y|$, *hamming distance* = minimum # substitutions needed to turn one into the other

For X, Y , *edit distance* = minimum # edits (substitutions, insertions, deletions) needed to turn one into the other

If $|X| = |Y|$ what can we say about the relationship between **editDistance**(X, Y) and **hammingDistance**(X, Y)?

$$\text{editDistance}(X, Y) \leq \text{hammingDistance}(X, Y)$$

Substructure optimality



Recurrence

α C

β A

$$\text{edist}(\alpha C, \beta A) = \min \begin{cases} \text{edist}(\alpha, \beta) + 1 \\ \text{edist}(\alpha C, \beta) + 1 \\ \text{edist}(\alpha, \beta A) + 1 \end{cases}$$

Recurrence general

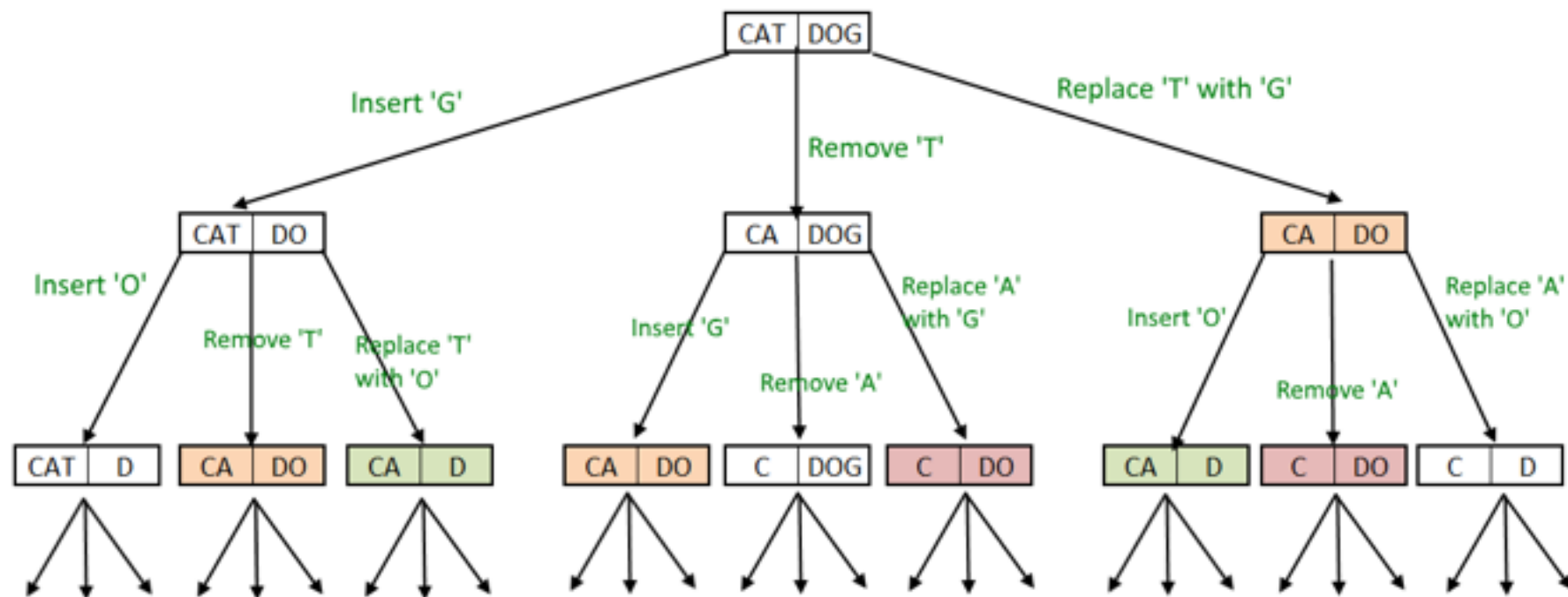
α x

β y

$$\text{edist}(\alpha x, \beta y) = \min \begin{cases} \text{edist}(\alpha, \beta) + \delta(x, y) \\ \text{edist}(\alpha x, \beta) + 1 \\ \text{edist}(\alpha, \beta y) + 1 \end{cases}$$

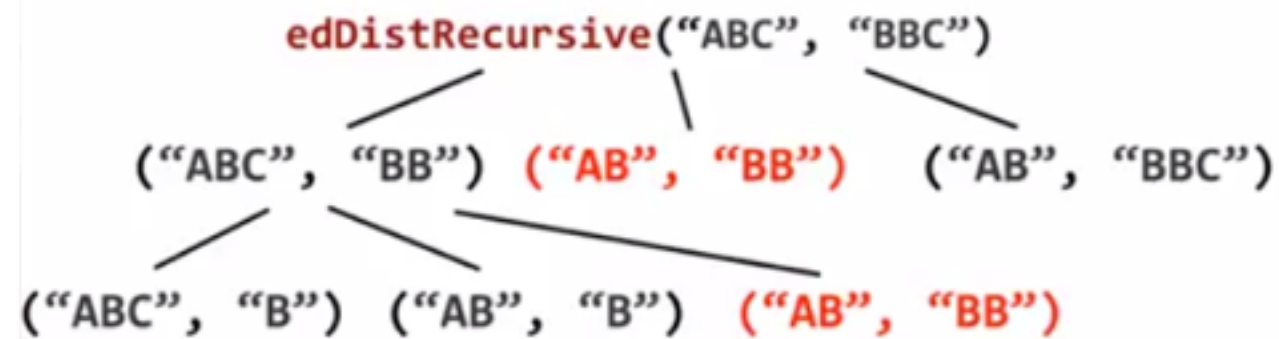
$\delta(x, y) = 0$ if $x = y$, or 1 otherwise

Recursion tree for an example



Recursive function

```
>>> import datetime as d
>>> st = d.datetime.now(); \
... edDistRecursive("Shakespeare", "shake spear"); \
... print (d.datetime.now()-st).total_seconds()
3
31.498284
```



Repetitions

```
n = 0
def edDistRecursive(a, b):
    global n
    if len(a) == 0:
        return len(a)
    if len(b) == 0:
        return len(b)
    if a == 'Shake' and b == 'shake':
        n += 1
    delt = 1 if a[-1] != b[-1] else 0
    return min(edDistRecursive(a[:-1], b[:-1]) + delt,
               edDistRecursive(a[:-1], b) + 1,
               edDistRecursive(a, b[:-1]) + 1)
```

```
>>> edDistRecursive("Shakespeare", "shake spear")
3
>>> n
8989
```

Instance

		Y							
		ε	G	C	T	A	T	A	C
X	ε								
	G								
	C								
	G								
	T								
	A								
	T								
	G								
	C								

$$\boxed{\text{edist}(\alpha x, \beta y)} = \min \begin{cases} \text{edist}(\alpha, \beta) + \delta(x, y) \\ \text{edist}(\alpha x, \beta) + 1 \\ \text{edist}(\alpha, \beta y) + 1 \end{cases}$$

Instance

		Y							
		ε	G	C	T	A	T	A	C
X	ε								
	G								
	C								
	G								
	T								
	A								
	T								
	G								
	C								

$$\text{edist}(\alpha x, \beta y) = \min \begin{cases} \text{edist}(\alpha, \beta) + \delta(x, y) \\ \text{edist}(\alpha x, \beta) + 1 \\ \text{edist}(\alpha, \beta y) + 1 \end{cases}$$

Instance

		Y							
		ε	G	C	T	A	T	A	C
X	ε	0	1	2	3				
	G	1	0	1	2				
	C	2	1	0	1				
	G	3	1	1	1				
	T								
	A								
	T								
	G								
	C								

$$\text{edist}(\alpha x, \beta y) = \min \begin{cases} \text{edist}(\alpha, \beta) + \delta(x, y) = 0 + 1 = 1 \\ \text{edist}(\alpha x, \beta) + 1 = 1 + 1 = 2 \\ \text{edist}(\alpha, \beta y) + 1 = 1 + 1 = 2 \end{cases}$$

End